

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Computer Fundamental (CS-402)
End sem-I

Total Marks: 50

Date: 20.12.2024@10am

Time: 3hr

All Compulsory

1. Write note on

- a. Parity checker and generator
- b. The need of minimization in a logic circuit
- c. Combinational and sequential circuits
- d. Any two ID digital logic families
- e. Edge triggering and Level triggering

3*5=15

2. Represent decimal number 8620 in

- a. BCD
- b. Excess-3 code
- c. 2421
- d. Binary
- e. Hexadecimal

5

3. What will be the output signal f and g for logic circuit shown in fig 1 with following a, b and c input signals shown in Fig 2?

5

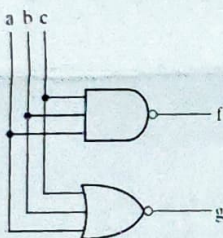


Fig 1 : Logic Circuit

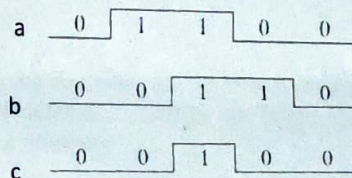


Fig 2 : Input Signal

4. Simplify the Boolean Function F together with don't care conditions d in (i) sum of product and (ii) Product of sum

$$F(w, x, y, z) = \Sigma(0, 1, 2, 3, 7, 8, 10)$$

$$d(w, x, y, z) = \Sigma(5, 6, 11, 15)$$

5

5. Implement a 3 bit magnitude comparator.

5

6. Implement the following functions with PLA

5

$$F1(A, B, C) = \Sigma(3, 5, 6, 7)$$

$$F2(A, B, C) = \Sigma(0, 2, 4, 7)$$

7. a) What is a flip flop? Explain the T-flip flop alongwith state table, characteristic table with characteristic equation.

b) Implement 3 bit step-up counter using T flip. Make state diagram, excitation table and logic circuit.

4+6